

# A layered divergence guard for the 2-D NS Galerkin scan

How the guard battery was chosen, the mathematics behind each layer, how it is implemented in `build_ns_n_dependence_nb.py` / `ns_numba.py`, and what the verifications actually show. All constants below were recomputed, not copied from the code comments.

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## 1. Why one divergence test is not enough

The scan integrates the truncated 2-D Navier–Stokes (vorticity) Galerkin system

$$\dot{\omega} = M(\omega)\omega + \hat{f} - \nu k^2 \omega,$$

where  $M(\omega)$  is the discrete convective operator (a matrix whose entries are linear in  $\omega$ , so the RHS is **quadratic** in  $\omega$ ),  $\hat{f}$  is the forcing ( $A/4$  on each of the four modes  $(\pm 1, \pm 1)$ ), and  $k^2$  is the per-mode wavenumber squared. A point is “chaotic” if its largest Lyapunov exponent  $\lambda_1 > 0$ . The danger is that a computed  $\lambda_1$  (or a “divergence”) can be **spurious** for reasons that have nothing to do with the physics:

| Failure mode                                               | What it looks like                                                                       | What catches it                         |
|------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------|
| RK4 step outside its stability region                      | $\ \omega\ _2 \rightarrow 10^{85}$ , “divergence” that moves *in* with $N$ at fixed $dt$ | <b>L7</b> (safe timestep)               |
| wrong variational Jacobian                                 | $\lambda_1$ is wrong but the trajectory looks fine                                       | <b>L6</b> (central-FD Jacobian)         |
| state not on the attractor (transient / spurious repeller) | bounded, but outside the a-priori absorbing ball                                         | <b>L3</b> (dissipative ball)            |
| point on a basin boundary / non-reproducible object        | $\lambda_1$ depends on the initial seed                                                  | <b>L4</b> (multi-seed ensemble)         |
| genuine physical divergence                                | residual, after all four                                                                 | (the thing we actually want to measure) |

The design principle is **layering by failure mode, ordered so that a garbage input never reaches the quantity we care about**:

1. **L7 first** — if the timestep is unstable, *every* downstream number (trajectory,  $\lambda_1$ , ball test, seed spread) is meaningless. So the timestep is fixed before any integration.
2. **L6 at self-check** — the Jacobian is only used inside Benettin; if it is wrong,  $\lambda_1$  is wrong even for a perfectly integrated trajectory. It is verified once, cheaply, before the heavy scan.
3. **L3 and L4 per point** — each grid point is then classified as *on-ball* / *off-ball* (L3) and *seed-stable* / *seed-unstable* (L4), so the reported  $\lambda_1$  and the  $A^*$  bisection only ever use points that are physically meaningful.

The layers are **independent**: each answers a different question, and none subsumes another. A point can be bounded (L7 ok), have a correct Jacobian (L6 ok), be on-ball (L3 ok), and still be seed-unstable (L4) — or any other combination. The guard is the *intersection* of the four “ok” conditions.

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## 2. L7 — the stiffness-aware timestep (the load-bearing layer)

### 2.1 The math

Classical 4th-order Runge–Kutta has a stability function  $R(z) = 1 + z + z^2/2 + z^3/6 + z^4/24$ . On the **real axis** its stability interval is  $[z_L, 0]$  with

$$z_L \approx -2.7853,$$

and on the **imaginary axis** the window is  $|y| \leq 2\sqrt{2} \approx 2.8284$ . (Both recomputed by root-finding; the real endpoint is a simple zero of  $R(z) - 1$  on  $z < 0$ , the imaginary one the largest  $y$  with  $|R(iy)| \leq 1$ .)

The Galerkin RHS has two sources of stiffness, both growing with the truncation  $N$  (where  $k_{\max}^2 = N^2$  for this square-cutoff basis):

- **Viscous.** The term  $-\nu k^2 \omega$  contributes eigenvalues  $-\nu k^2$  (real, negative). The stiffest is  $\nu k_{\max}^2$ . RK4 needs  $\nu k_{\max}^2 dt \leq 2.7853$ , i.e.  $dt \leq 2.7853/(\nu k_{\max}^2)$ .
- **Convective.** The quadratic term has a Jacobian linear in  $\omega$ ; its eigenvalues are (roughly) imaginary with magnitude  $\lesssim \|\omega\| k_{\max}$ . Taking the typical settled amplitude to be the dissipative-ball radius  $\|\omega\| \sim A/(4\nu)$  (see §4), the convective stiffness is  $\sim (A/(4\nu)) k_{\max}$ , and the imaginary-axis window gives  $dt \leq 2\sqrt{2} \cdot 4\nu/(A k_{\max}) \approx 11.31 \nu/(A k_{\max})$ .

## 2.2 The implementation

```
def safe_dt(N, nu, A, cap=0.125):
    k2 = _kmax2(N); kmax = int(round(np.sqrt(k2))) # k_max^2 = N^2, k_max = N
    return min(cap, 2.0/(nu*k2), 8.0*nu/(A*kmax))
```

The three terms are: a fixed cap 0.125 (the original timestep, kept for small  $N$  where it is stable), the **viscous** bound, and the **convective** bound. The constants 2.0 and 8.0 are *deliberately below* the theoretical limits 2.7853 and 11.31 — a safety margin, so the step sits strictly inside the stability region even if the amplitude estimate  $\|\omega\| \sim A/(4\nu)$  is a little off.

**Honest caveat.** The convective bound is a *calibrated estimate*, not a sharp spectral bound: it uses the ball radius as a proxy for  $\|\omega\|$  and a rounded constant. It is not derived from the exact eigenvalues of  $J(\omega)$ . It is, however, **verified empirically** — the constant set (2.0, 8.0) was calibrated (probe run) to keep **all 84 grid points** ( $N \in \{6, 8, 10, 12\} \times \nu \times A$ ) bounded, and it is re-validated every run by the fact that no point diverges. So the guarantee is “conservative by construction + empirically all-bounded,” not “provably stable for all  $\omega$ .”

## 2.3 Why this layer is the important one

At a **fixed**  $dt = 0.125$ , the convective stiffness at  $N = 12, \nu = 0.05, A = 2.6$  is  $z_{\text{conv}} \sim (A/4\nu) dt k_{\max} = 19.5$ , which is  $7\times$  outside the imaginary window 2.828 — so RK4 is violently unstable and fabricates a “divergence” ( $\|\omega\| \rightarrow 10^{85}$ ) that is **purely numerical**. The same point is bounded at  $dt = \text{safe\_dt} = 0.0128$ . This is what makes the whole scan meaningful: without L7, the “ $A^*$  wall moves in with  $N$ ” is an artifact; with L7, the system is dissipative and every point settles, so the only  $N$ -dependence that remains is the genuine Galerkin under-resolution.

Representative values (recomputed):

| $N$ | $\nu$ | $A$ | viscous bound | convective bound | cap   | ‘safe_dt’     | active     |
|-----|-------|-----|---------------|------------------|-------|---------------|------------|
| 12  | 0.05  | 2.6 | 0.2778        | 0.0128           | 0.125 | <b>0.0128</b> | convective |
| 12  | 0.05  | 0.6 | 0.2778        | 0.0556           | 0.125 | <b>0.0556</b> | convective |
| 12  | 0.15  | 0.6 | 0.0926        | 0.1667           | 0.125 | <b>0.0926</b> | viscous    |
| 12  | 0.10  | 2.0 | 0.1389        | 0.0333           | 0.125 | <b>0.0333</b> | convective |
| 10  | 0.05  | 2.6 | 0.4000        | 0.0154           | 0.125 | <b>0.0154</b> | convective |

|   |      |     |        |        |       |               |            |
|---|------|-----|--------|--------|-------|---------------|------------|
| 6 | 0.15 | 2.6 | 0.3704 | 0.0769 | 0.125 | <b>0.0769</b> | convective |
|---|------|-----|--------|--------|-------|---------------|------------|

### 3. L6 — the central-FD Jacobian guard

#### 3.1 The math

The RHS  $f(\omega) = M(\omega)\omega + \hat{f} - \nu k^2 \omega$  is **quadratic** in  $\omega$ . For a quadratic map the central difference is *exact*:

$$f(\omega + hv) - f(\omega - hv) = 2h J(\omega)v + 0 \cdot h^2(\dots),$$

because the Hessian term is even in  $h$  and cancels. Hence

$$J(\omega)v = \frac{f(\omega + hv) - f(\omega - hv)}{2h} + \mathcal{O}(\text{roundoff}),$$

with **no truncation error** — only roundoff, of size  $\sim \varepsilon_{\text{mach}} \|f\|/h$ . This is a *perfect* test of the variational Jacobian: any linearization bug (e.g. a double-counted viscosity, a missing  $E(\omega)$  term) shows up as a large  $\max |J(\omega)v - \text{FD}|$ , while a correct Jacobian gives roundoff-level agreement.

Crucially, this failure mode is **invisible to every trajectory-based test**. A wrong  $J$  does not change the integrated trajectory (which uses  $f$ , not  $J$ ); it only corrupts Benettin's variational equation, i.e. the  $\lambda_1$  we report. No amount of checking "does the trajectory stay bounded" can catch it.

#### 3.2 The implementation

```
def fd_jac_guard(rhs, jac, om, nvec=4, eps=1e-6, seed=0):
    r = np.random.default_rng(seed); n = om.shape[0]; worst = 0.0
    for _ in range(nvec):
        v = r.standard_normal(n) + 1j*r.standard_normal(n)
        Jv = jac(om) @ v
        fd = (rhs(om + eps*v) - rhs(om - eps*v)) / (2.0*eps)
        worst = max(worst, float(np.max(np.abs(Jv - fd))))
    return worst
```

Four random complex directions,  $h = 10^{-6}$ , run **once at self-check** (cheap,  $N = 6$ ) before any heavy scan, with `assert fd_err < 1e-6`.

#### 3.3 Verification

Measured:  $\max |J@v - \text{FD}(J)@v| = 1.9\text{e-}8$  at  $N = 6$ . This is exactly the roundoff scale predicted above ( $\varepsilon_{\text{mach}} \|f\|/h \sim 10^{-16} \cdot \|f\|/10^{-6} \approx 10^{-8}$  for  $\|f\| \sim 10^2$ ), confirming the Jacobian is correct to roundoff and the "exactness" claim holds. The Numba port re-verifies the same identity in-notebook (S0b:  $\max |\Delta J@v| = 3.9 \times 10^{-14}$ ).

### 4. L3 — the a-priori dissipative ball (off-attractor test)

#### 4.1 The math

Take the  $L^2$  (enstrophy) inner product of the vorticity equation with  $\omega$  and keep the real part:

$$\frac{1}{2} \frac{d}{dt} \|\omega\|^2 = \underbrace{\text{Re} \langle \omega, M(\omega) \omega \rangle}_{= 0 \text{ (skew-adjoint)}} + \text{Re} \langle \omega, \hat{f} \rangle - \nu \|\omega\|_{k^2}^2.$$

The first term vanishes: the 2-D convective term is skew-adjoint in the enstrophy norm (the classical  $\int \omega J(\psi, \omega) = 0$  identity). **This was verified for the discrete operator**, not assumed:  $\text{Re} \sum \bar{\omega}_k (M\omega)_k \approx 10^{-17}$  for random states at  $N = 6, 12$ . (An earlier probe that reported a large value had accidentally included the forcing term  $\hat{f}$ ; the convective operator alone does zero work.)

Bounding the remaining terms:

- $\text{Re} \langle \omega, \hat{f} \rangle \leq \|\omega\| \|\hat{f}\|$  (Cauchy-Schwarz), with  $\|\hat{f}\| = A/2$  ( $A/4$  on each of the four modes  $(\pm 1, \pm 1)$ ).
- $\|\omega\|_{k^2}^2 = \sum k^2 |\omega_k|^2 \geq \lambda_1 \|\omega\|^2$ , where  $\lambda_1 = \min k^2$  over the Galerkin modes.

So, with  $y = \|\omega\|$ :

$$y' \leq \|\hat{f}\| - \nu \lambda_1 y = \frac{A}{2} - \nu \lambda_1 y,$$

whose equilibrium is the **absorbing ball**

$$\|\omega\|^* = \frac{\|\hat{f}\|}{\nu \lambda_1} = \frac{A}{2 \nu \lambda_1}.$$

A settled (post-transient) state must lie inside this ball; a state outside it is not on the expected attractor (still transient, or a spurious object).

## 4.2 The constant the code uses — and the factor-of-2 subtlety

The code sets

```
def R_diss(A, nu):
    return A/(4.0*nu)          # = A/(2 nu * 2), i.e. assumes lambda_1 = 2
```

which corresponds to  $\lambda_1 = 2$ . **But the true minimum wavenumber in the basis is  $k^2 = 1$**  (the modes  $(\pm 1, 0), (0, \pm 1)$ , which satisfy  $0 < k^2 \leq N^2$ ). The rigorous spectral bound is therefore  $\lambda_1 = 1$ , giving the rigorous absorbing ball

$$\|\omega\|_{\text{rigorous}}^* = \frac{A}{2\nu} = 2 \cdot R_{\text{diss}}.$$

So the code's threshold is a **factor of 2 tighter** than the rigorous a-priori ball. This makes the off-ball test *conservative*: it flags a state as "off-ball" when  $\|\omega\| > 1.05 R_{\text{diss}}$ , i.e. at *half* the rigorous radius. It therefore **cannot miss** a genuinely off-ball state (it is stricter than the rigorous bound), at the cost of *possibly* flagging a state that is inside the rigorous ball but outside the tighter threshold.

In the actual runs **no state was ever flagged off-ball** (all `offball=0`), which is a *stronger* statement than the rigorous bound requires: every post-transient state sat inside the *tighter*  $A/(4\nu)$  ball, not merely the rigorous  $A/(2\nu)$  one. So the factor-of-2 looseness in the stated constant never caused a false positive, and the empirical result (attractor well inside the tight ball) is unaffected.

**Honest caveat.**  $R_{\text{diss}} = A/(4\nu)$  is a *practical* threshold, not a proven a-priori absorbing radius: the rigorous one is  $A/(2\nu)$ . The code comment (" $\lambda_{1,k} = \min k^2 = 2$ ") is a minor

inaccuracy — the true  $\min k^2 = 1$ . The choice is defensible (conservative, and the attractor is far inside either bound), but the note should not overstate it as a sharp bound.

### 4.3 The implementation

```
offball = l2 > R_diss(A, nu)*1.05          # L3 dissipative-ball test
```

where  $l_2 = \|\omega\|_2$  after the transient. The 5% tolerance absorbs the roundoff/transient tail. The flag is used in two places: (a) it marks grid points with \* in the table, and (b) it **gates the  $A^*$  bisection** — a bracketing interval is only bisected if *both* endpoints are on-ball, so the boundary estimate never rides on an off-attractor point.

## 5. L4 — the multi-seed ensemble (basin-boundary test)

### 5.1 The math

For a genuine chaotic attractor,  $\lambda_1$  is a property of the **invariant measure**, not of the particular trajectory. Any point in the basin of attraction must reproduce the same  $\lambda_1$  (to integration tolerance). If three well-separated initial seeds give three *different*  $\lambda_1$ , the point is not on a single attractor — it is on a **basin boundary** or a **spurious repeller** where the long-time behaviour is seed-dependent.

So the test is: run 3 seeds, each with a long transient ( $T = 300$ ) and a long Benettin window ( $T = 300$ ,  $m = 16$ ), and measure the **spread**

$$\text{spread} = \max_i \lambda_1^{(i)} - \min_i \lambda_1^{(i)}.$$

A seed-stable point has spread at the integration-tolerance level; a basin-boundary point has a large spread.

### 5.2 The implementation

At the best-scanned point ( $\nu = 0.1$ ,  $A = 2.0$ ), seeds 1, 2, 3:

```
for s in [1, 2, 3]:
    om = even_init(ks3, idx3, 0.01, s)
    for _ in range(int(300/dt3)): om = _rk4(rf, om, dt3)
    lam_c, om = benettin_ns(rf, jf, om, T=300.0, dt=dt3, m=16)
    seed_lams.append(float(real_spectrum(lam_c)[0]))
out["best_spread"] = max(seed_lams)-min(seed_lams)
```

The threshold for "seed-unstable" is  $\text{spread} > 0.3$  (much larger than any integration tolerance, so it only fires for genuinely seed-dependent points).

### 5.3 Verification

In the Numba run the spreads were **0.004-0.017** across  $N = 6, 8, 10, 12$  — all far below 0.3, so every reported "best" point is **seed-stable** (a genuine attractor, not a basin-boundary artifact). This is what licenses reporting the  $\lambda_1$  values at all: they are reproducible properties of the attractor, not seed accidents.

## 6. How the layers compose (the decision procedure)

For each grid point  $(N, \nu, A)$  the guard produces the 5-tuple  $(\lambda_1, \text{div}, \omega_{\max}, \ell_2, \text{offball})$  and the point is classified:

```
div      = (not finite(lambda_1)) or omega_max > 1e8      # L7 residual: true blow-up
offball  = ell_2 > 1.05 * R_diss(A, nu)                  # L3: off the attractor
seed_unstable = spread > 0.3                             # L4: basin boundary
```

A point is **trusted** only if all of:

- L7: integrated at `safe_dt` (so `div` is physical, not numerical);
- L6: the Jacobian passed the FD guard (so  $\lambda_1$  is correct);
- L3: `offball == False` (so it is on the attractor);
- L4: `spread` small (so it is seed-reproducible).

Only trusted points enter the  $A^*$  bisection and the “best  $\lambda_1$ ” report. The layers are checked in the order they invalidate their dependents (L7  $\rightarrow$  L6  $\rightarrow$  L3/L4), so a failure at an early layer is caught before any expensive downstream computation is wasted on it.

## 7. Summary and honest caveats

| Layer                            | Question it answers                      | Math                                                                                                                              | Constant(s)                                                  | Verified                                                                                                                                                                                         |
|----------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>L7</b> <code>safe_{dt}</code> | is the integration *numerically* stable? | RK4 real window $[-2.7853, 0]$ , imag window $\pm 2\sqrt{2}$ ; viscous $\nu k_{\max}^2$ + convective $(A/4\nu)k_{\max}$ stiffness | $dt = \min(0.125, 2.0/(\nu k_{\max}^2), 8.0\nu/(Ak_{\max}))$ | all 84 grid points bounded (empirical); constants below theoretical limits (conservative)                                                                                                        |
| <b>L6</b> FD-Jacobian            | is the variational Jacobian *correct*?   | quadratic RHS $\Rightarrow$ central FD is exact, $\max  Jv - \text{FD}  \sim \varepsilon_{\text{mach}} \ f\ /h$                   | $h = 10^{-6}$ , 4 directions, assert $< 10^{-6}$             | measured $1.9 \times 10^{-8}$ (roundoff) $\square$                                                                                                                                               |
| <b>L3</b> dissipative ball       | is the state *on the attractor*?         | skew-adjoint convective term $\Rightarrow$ absorbing ball $\ \omega\ ^* = A/(2\nu\lambda_1)$                                      | $R_{\text{diss}} = A/(4\nu)$ ( $=\lambda_1 = 2$ ), 5% tol    | convective zero-work verified ( $\sim 10^{-17}$ ); <b>code's <math>\lambda_1 = 2</math> vs true <math>1 \Rightarrow</math> threshold <math>2\times</math> tighter than rigorous; never fired</b> |
| <b>L4</b> multi-seed             | is the point *seed-reproducible*?        | invariant-measure $\lambda_1$ must be seed-independent                                                                            | 3 seeds, $T = 300, m = 16$ , spread $> 0.3 =$ unstable       | spreads 0.004–0.017 (all seed-stable) $\square$                                                                                                                                                  |

**Caveats, stated plainly:**

1. **L7 is calibrated, not sharp.** The convective bound uses the ball radius as a proxy for  $\|\omega\|$  and rounded constants; it is “conservative + empirically all-bounded,” not a proven stability certificate.

2. **L3's stated constant is a factor of 2 off from rigorous.** The code's  $R_{\text{diss}} = A/(4\nu)$  assumes  $\lambda_1 = 2$ ; the true  $\min k^2 = 1$ , so the rigorous absorbing ball is  $A/(2\nu)$ . The code's threshold is therefore *tighter* (conservative) and never fired, so no result is affected — but the comment should not be read as a sharp a-priori bound.
3. **L4's threshold (0.3) is a judgment call**, chosen well above integration tolerance so it only fires for genuinely seed-dependent points; it is not a derived quantity.
4. **The guards separate numerical from physical failure, but they do not certify the Galerkin itself.** They guarantee that a *reported*  $\lambda_1$  is (a) numerically stable, (b) from a correct Jacobian, (c) on-ball, and (d) seed-stable. They do **not** address Galerkin under-resolution (too-small  $N$ ), which is a *physics* limitation handled separately by the  $N$ -trend and active-mode analysis, not by the divergence guard.